

EVALUATING TREATMENT OPTIONS FOR CHRONIC KIDNEY DISEASE STAGE 5 USING A MARKOV DECISION PROCESS

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ABSTRACT

Chronic Kidney Disease (CKD) affects approximately 1 in 7 U.S. adults [5], many of whom remain unaware of their condition. The disease progresses gradually, ultimately reaching a stage in which the kidneys lose their ability to effectively filter waste and excess fluid from the bloodstream. A Markov Decision Process (MDP) can be used to model the progression of CKD, providing insight into the disease's impact and helping identify optimal treatment strategies over time.

MODEL

This model assumes that a patient is currently in CKD stage 5. We use a Markov Decision Process (MDP) to model the disease.

- s_0 : the initial state
- $\text{ACTION}(s)$: a set of actions in each state
- $P(s'|s, a)$: transition model
- $R(s)$: reward function
- π : solution, which is a policy

In our model, s_0 is CKD stage 5, and the patient must decide on a treatment. The options are conservative care (CC), transplant, or dialysis.

Regardless of the choice of treatment, there are several conditions that the patient is at risk for, including stroke, stroke leading to disability, myocardial infarction (MI), MI leading to congestive heart failure, and death.

BACKGROUND

CKD progresses through five distinct stages, each indicating a greater loss of kidney function. The following terms are necessary to understand this project:

- Quality of Life (QoL): ranges from 0.0 to 1.0
- Quality Adjusted Life Years (QALY)
- Dialysis: Catheter vs Fistula
- Conservative Care (CC): not transplant or dialysis

METHOD 1

Value Iteration: The MDP we have has an infinite horizon (not a fixed time), so the optimal policy is stationary (depends only on the current state). The model will always end in a terminal state. We use additive rewards where the discount factor, $\gamma = 1$. Utilities are defined based on the QoL. To get the expected utility:

$$U_s^\pi = E\left[\sum_{t=0}^{\infty} \gamma^t R(S_t)\right]$$

To get the optimal policy:

$$\pi_s^* = \arg \max_{\pi} U^\pi(s)$$

Then use the Bellman equation to iteratively get the next transition:

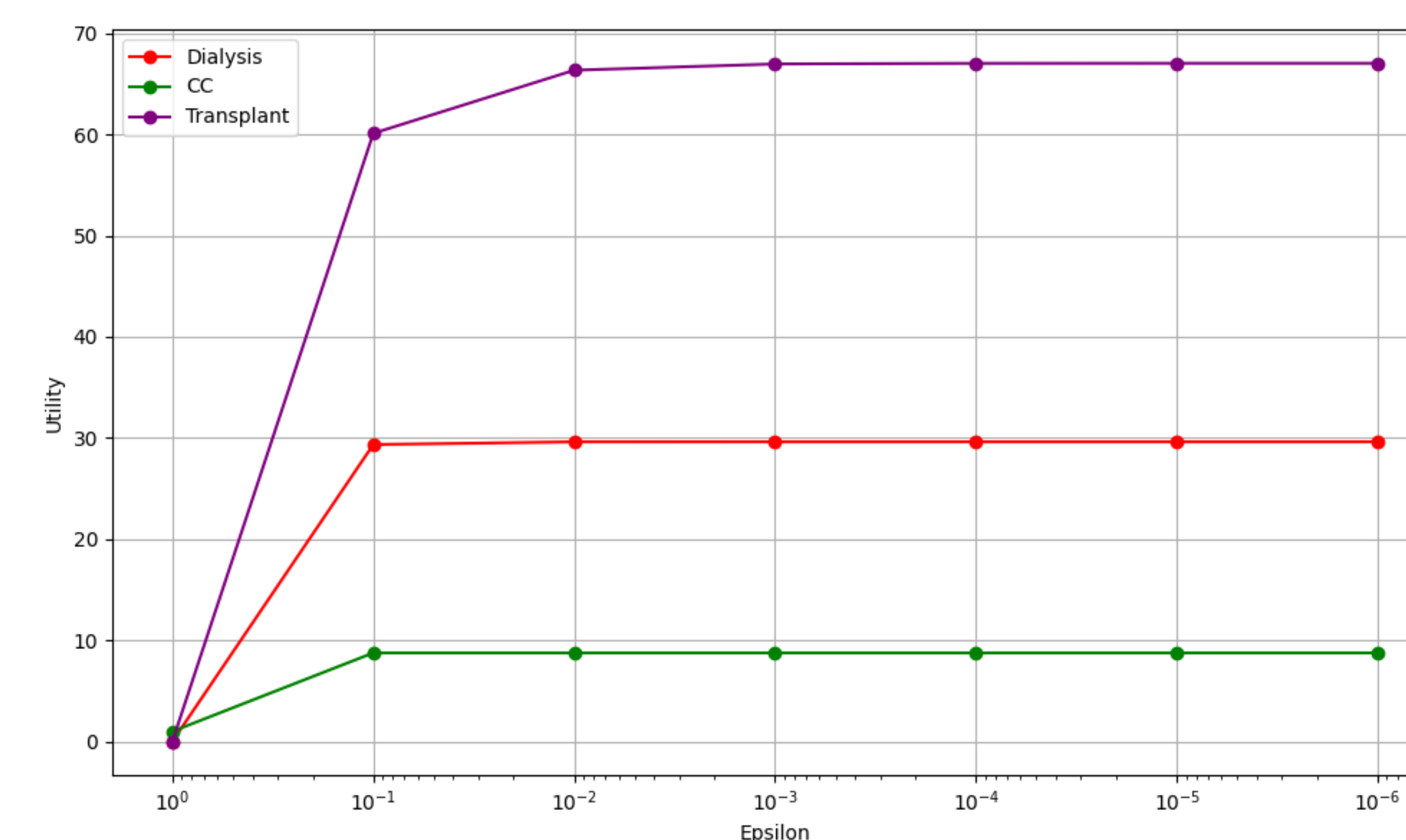
$$U_{t+1}(s) \leftarrow R(s) + \gamma \max_{a \in A(s)} \sum_{s'} P(s' | s, a) U_t(s')$$

Keep applying the updates until the greatest change is lower than some ϵ .

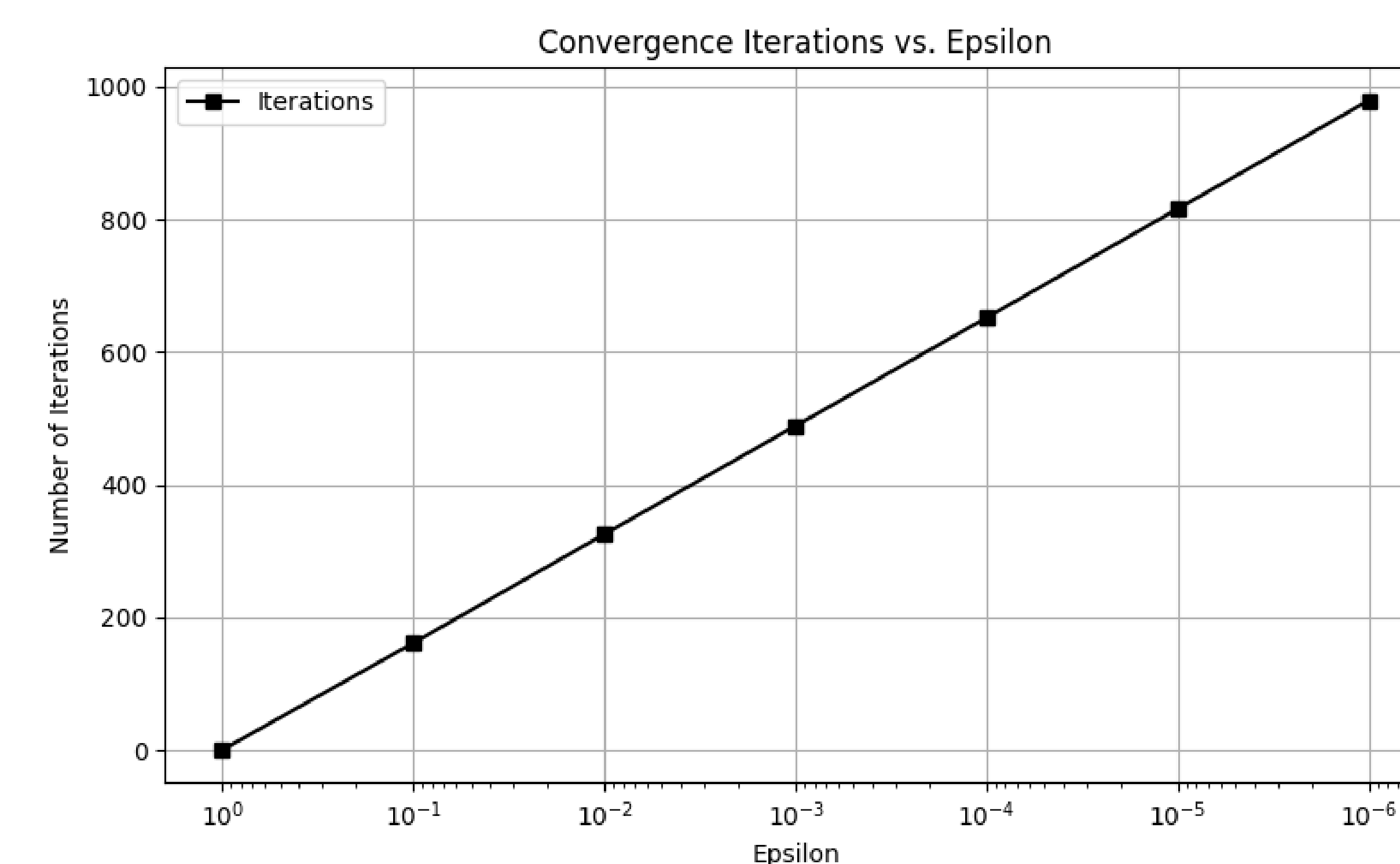
VALUE ITERATION RESULTS

Metric	Cumulative Rewards	Normalized	Adjusted
Dialysis	29.62	0.442	0.392
CC	8.75	0.130	0.116
Transplant	67.04	1.000	0.887
Fistula	29.62	0.442	0.392
Dialysis w/ fistula	25.84	0.385	0.342

The first column is the cumulative rewards, the second is normalized to one, and the third is normalized to the transplant value from method 2. The best policy is to choose transplant, and if you are in dialysis choose fistula.



Utility vs Epsilon



Iterations vs Epsilon

The plot on the left shows that the utility values converge at low values of epsilon. This means that the epsilon value of $10E - 4$ was small enough to result in accurate results.

The second plot shows the utility and the number of iterations it takes for the MDP to converge. Interesting, there is a linear relationship between the two.

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METHOD 2

We can also solve the MDP through simulation. Start in s_0 , which is CKD stage 5, and from there, we apply the transition by taking a random weighted choice to get to the next state. If we reach a decision node then we randomly make the choice. This approach allows for more analysis such as the number of months each patient lives after being diagnosed with CKD 5, the average QoL, and the average QALY.

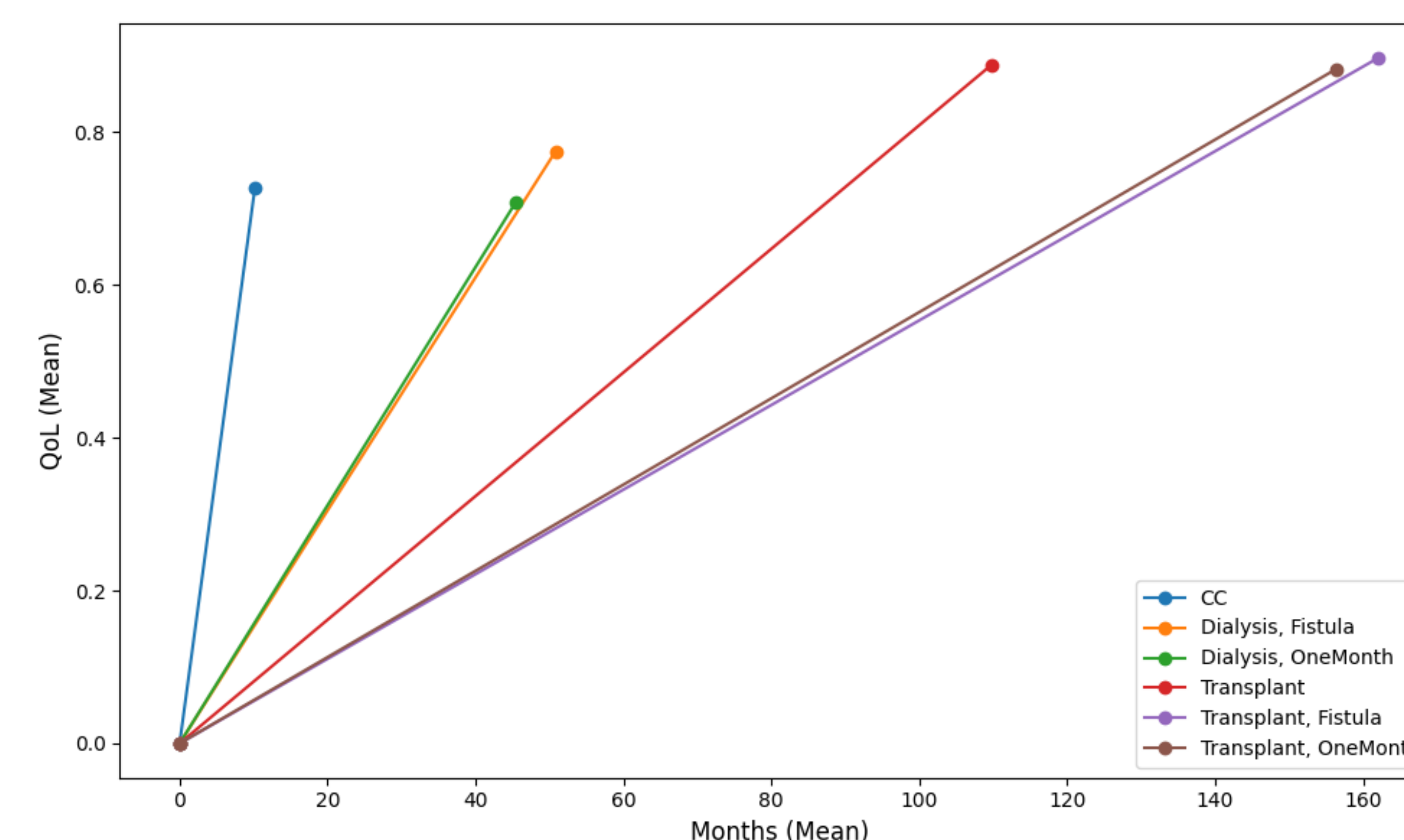
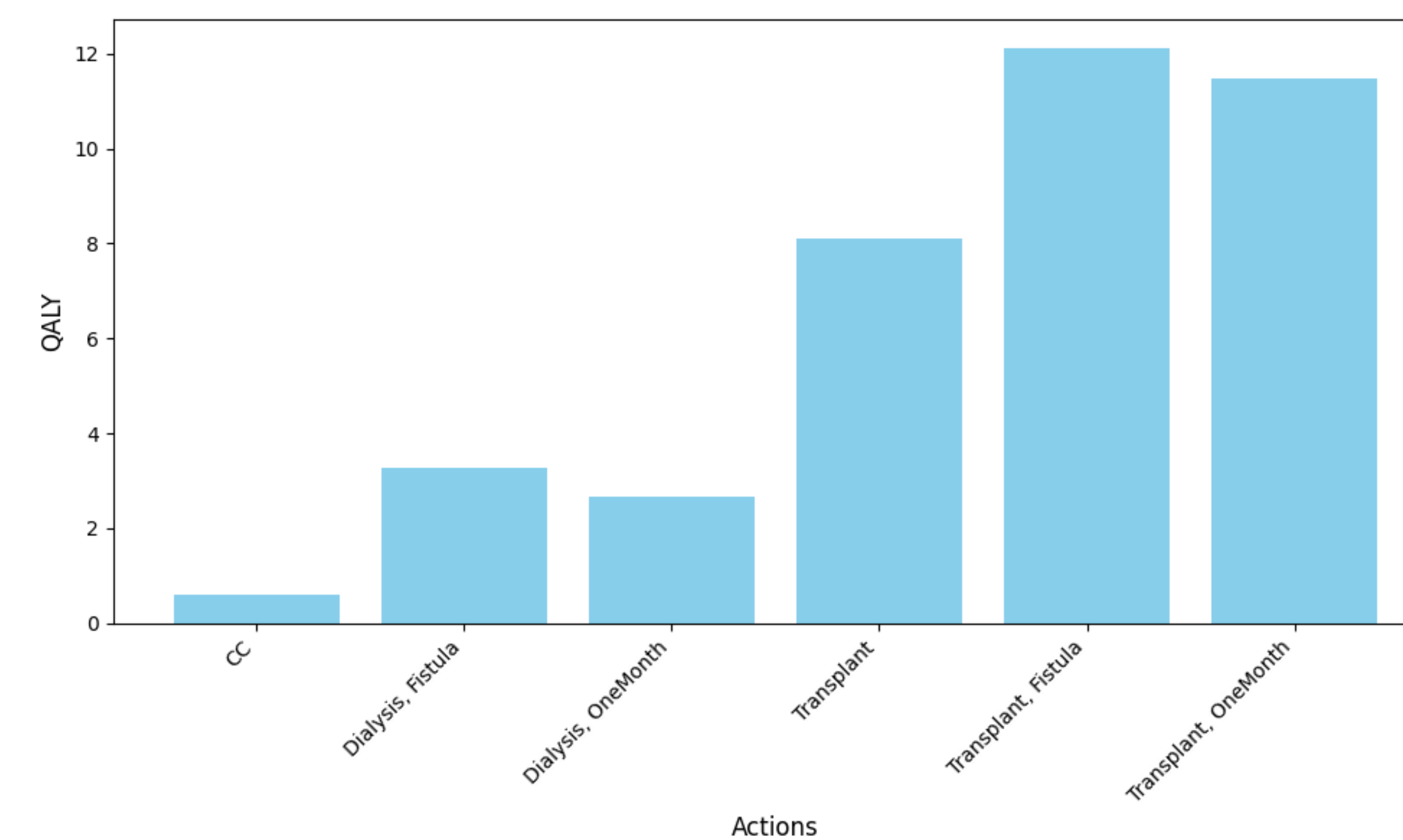
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SIMULATION RESULTS

Decision	Months (Mean)	Months (SD)	QoL (Mean)	QoL (SD)	QALY
CC	10.164369	9.566915	0.726293	0.272583	0.615192
Dialysis, Fistula	50.832281	50.297105	0.774744	0.130685	3.281833
Dialysis, OneMonth	45.413938	45.875516	0.707516	0.170910	2.677592
Transplant	109.760794	112.305906	0.887768	0.162923	8.120172
Transplant, Fistula	162.093939	122.189669	0.897094	0.035479	12.117785
Transplant, OneMonth	156.330637	118.209652	0.882115	0.047723	11.491801

Table 1: Results from simulation with 100,000 Note: OneMonth means regular dialysis with no fistula



The graph on the right clearly shows that the most QALY is from transplant, and then fistula. This means that if you have a transplant, and then it fails, you should then go onto dialysis with a fistula. People who outlive their transplant are rare, so it makes sense they would live longer if they outlive their transplant and then go on dialysis. Therefore, our results match with method 1, you should choose transplant, and if you need dialysis you should choose fistula.

The second graph shows QoL vs the lifespan of an individual. This is just another way to represent QALY. It is interesting how the CC has a fairly high QoL, but a very short lifespan, resulting in its low QALY. The model seems to reward living longer, but some patients may prioritize short-term QoL over the long-term QALY. If we wanted to model this, we could add in a discount rate, similar to the γ value in method 1, which may alter our results slightly, depending on the value of γ .